VF Asia Footwear Laboratory Testing Manual

Test Method for Midsole Sampling	Test method # VF-H03 Page 1 of 4
Date Issued : May 22 , 2017 **Revision # : 1** **Reason for Revision: Changed the thickness of resiliency, skin on / off for some parameters and location of measurement.**	**Created by : Nila Babia Erazo**

PURPOSE AND SCOPE: These techniques must be used when testing midsoles or unit-soles, sample must be taken from the described locations. For compression or injection-molded mid-soles, testing must be performed with the specified skin-on and skin-off designations.

REFERENCES:

MATERIALS: Foamed mid-soles and unit-soles.

SAMPLING:

Test samples for midsole and unit-sole testing should be as follows:

A) Molded polyurethane Mid-soles or Unit-soles

Test	Thickness (mm)	Skin	Measurement Location
Durometer	10	On	Heel / Forefoot
Specific Gravity	10	On	Heel / Forefoot
Tensile	3.0 ± 1	On (can be 1 side off)	Forefoot
Elongation	3.0 ± 1	On (can be 1 side off)	Forefoot
Die- C tear	3.0 ± 1	On (can be 1 side off)	Forefoot
Split Test	10	On can be Off	Heel / Forefoot
Compression set	10	On	Heel
Ross Flex	5	On	Forefoot
Resilience	12.5	On	Heel / Forefoot
Din Abrasion	10	On	Heel / Forefoot
Shrinkage	Based on unit- sole	On	Complete unit-sole
Light Discoloration	n/a	On	Exposed side
Water absorption	Based on unit- sole	On	Complete unit-sole

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B) Compression or injection- molded EVA Midsoles or 3DMF Unit-soles :

Test	Thickness (mm)	Skin	Measurement Location
Durometer	10	On	Heel / Forefoot
Specific Gravity	10	On	Heel / Forefoot
Tensile	3.0 ± 1	On (can be 1 side off)	Forefoot
Elongation	3.0 ± 1	On (can be 1 side off)	Forefoot
Die- C tear	3.0 ± 1	On (can be 1 side off)	Forefoot
Split Test	10	On can be Off	Heel / Forefoot
Compression set	10	On	Heel
Ross Flex	5	On	Forefoot
Resilience	12.5	On	Heel / Forefoot
Din Abrasion	10	On	Heel / Forefoot
Shrinkage	Based on unit- sole	On	Complete unit-sole
Light Discoloration	n/a	On	Exposed side
Water absorption	Based on unit- sole	On	Complete unit-sole

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C) Die-cut EVA midsoles

Test	Thickness (mm)
Durometer	10
Specific Gravity	10
Tensile	3.0 ± 1
Elongation	3.0 ± 1
Die- C tear	3.0 ± 1
Split Test	10
Compression set	10
Ross Flex	5
Resilience	12.5
Shrinkage	10
Light Discoloration	n/a
Water absorption	n/a

Notes:

1. If test plaques are being used, sample can be taken from any location, but must be of the proper thickness and “skin-on/ skin-off” configuration.

2. Specific Gravity for soft polyurethane foams is measured using the water displacement technique

3. The “split tear line” must be located and maintained at the midpoint of the foam’s thickness.

4. When performing the Ross flex test, the “skin-on” side must be facing upward.

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5. When performing the UV-light discoloration test, the sample must be taken from the side of the midsole sample (with skin-on), and the side must be facing upward in the UV-light chamber.

